

Mostafa Riazi | CV

 Zimmer 14, Universitätsstraße 98-B, 70569 Stuttgart
 +49-15756601592 •  mostafa.riazi1994@gmail.com
 [Google Scholar](#) •  [ResearchGate](#) •  [LinkedIn](#)

Skills

- ❖ Proficient in **Python** programming language and its data analysis libraries such as Numpy, Pandas, Matplotlib, Seaborn, etc.
- ❖ Expertise in **Deep learning models** implementation using **PyTorch** Python package for hydrological systems evaluation aiming at developing Long short-term memory (LSTM), Convolutional neural network (CNN), etc.
- ❖ Proficient in **Hydrologic Modeling** of the watersheds using a) the **SWAT** (Soil and Water Assessment Tool) model used for hydrological processes simulation, agricultural conservation practices, best management practices (BMPs), climate change modeling, water quality modeling, b) **IHACRES** (Identification of unit Hydrographs and Component flows from Rainfall, Evaporation and Streamflow data) model, and c) **machine learning** models, etc.
- ❖ Skilled in **Machine Learning** modeling using **Scikit-learn** for building models such as Tree-based algorithms, Boosting-based algorithms, K-Nearest Neighbors (KNN), Artificial neural networks (ANN), etc.
- ❖ Expertise in using **Geospatial Science** tools such as Geospatial Data Abstraction Library (GDAL), Rasterio, GeoPandas, and Whitebox for spatial data analysis and processing
- ❖ Proficient in using **Google Earth Engine (GEE)** for Image Analysis, Geospatial Analytics, Temporal Analysis, Machine Learning modeling implementation, Data Visualization, etc.
- ❖ Experienced in **climate change** assessment procedures including downscaling and bias correction procedures and dealing with data from NetCDF files

Research interests

- ❖ Hydrological/ Hydrogeological processes assessment (quality and quantity)
- ❖ Remote sensing (RS) applications
- ❖ Watershed management and BMPs
- ❖ Climate change impact evaluation
- ❖ Change detection using Machine Learning and RS
- ❖ Spatial and spatio-temporal natural hazards analysis

Education

- ❖ Ph.D. in Civil and Environmental Engineering
 - Institution/Place: University of Stuttgart, Stuttgart, Germany
 - Duration: Oct. 2025 - Present
- ❖ M.Sc. in Civil Engineering, Water Resources Management, GPA: 4/4 (i.e. 19.47/20)
 - Institution/Place: Islamic Azad University (IAU) of Khomeini Shahr Isfahan, Iran (#2 in Best University in Iran, US News)
 - Duration: Sep. 2019 -Jan. 2023
 - Thesis topic: Evaluation of a watershed hydrology using three types of hydrologic modeling approaches
- ❖ B.Sc. in Agricultural Engineering; Water
 - Institution/Place: Imam Khomeini International University, Qazvin, Iran
 - Duration: Sep. 2012 -Sep. 2016

Publications

- ❖ M.Riazi, et al., Enhancing flood susceptibility modeling using multi-temporal SAR images, CHIRPS data, and hybrid machine learning algorithms, Science of the Total Environment (IF=10.754), Feb 2023; ([DOI](#))
- ❖ M.Riazi, et al., Comparative assessment of advanced machine learning techniques for simulation of lake water level fluctuations based on different dimensionality reduction methods, Earth Science Informatics (IF=2.705), Jan, 2023; ([DOI](#))
- ❖ M.Riazi, et al., Assessing Groundwater Drought Vulnerability through Baseflow Separation and Index-Based Analysis under Climate Change Projections, Groundwater for Sustainable Development (IF=4.9), Apr 2024, ([DOI](#))
- ❖ M.Riazi, et al., Enhancing Rainfall-Runoff Simulation in Data Poor Watersheds: Integrating Remote Sensing Data and Hybrid Decomposition for Hydrologic Modelling, Water Resources Management (IF: 4.7), April 2025, ([DOI](#))

Language proficiency and scores

- ❖ English: Advanced level
 - TOEFL iBT: 104 (R: 30, L: 24, S: 25 , W: 25)
 - GRE General: 330 (Q: 170, V: 160, W: 4)
- ❖ German (Deutsch): Intermediate level
 - A2 Level (Carl Duisberg Centrum München)
- ❖ Persian: Native/Fluent

Experience

- ❖ Academic experience
 - Peer-reviewed scientific journals including: Scientific Reports, Remote Sensing in Earth Systems Sciences, Iranian Journal of Science and Technology, Transactions of Civil Engineering, Earth Science Informatics, Applied Computing and Geosciences, Acta Geophysica
 - Position: Reviewer (May 2023 - Present)
- ❖ Work experience
 - Ministry of Energy and Water (MoEW) of Afghanistan, Kabul, Afghanistan
 - Position: Water Resources Management Specialist (Sep.2020 - Jul. 2021)

Awards, Achievements, and Certificates

- ❖ Top student at Department of Mechanics, Civil Engineering, and Architecture at Islamic Azad University of Khomeini Shahr, 2022
- ❖ Top student at Department of Mechanics, Civil Engineering, and Architecture at Islamic Azad University of Khomeini Shahr, 2020
- ❖ Membership in the American Society of Civil Engineers (ASCE), 2023
- ❖ “Python for Everybody” course at School of Information, University of Michigan (Coursera), 2022
- ❖ “Machine Learning” course at Computer Science Department, Stanford University (Coursera), 2022
- ❖ “Deep Learning Specialization” course at Computer Science Department, Stanford University (Coursera), 2022
Neural Networks and Deep Learning Improving Deep Neural Networks Structuring Machine Learning Projects
Convolutional Neural Networks Sequence Models
- ❖ Elementary Workshop of The Soil Water Assessment Tool (SWAT) model at Tarbiat Modares University, Tehran, Iran, 2022
- ❖ Advanced Workshop of The Soil Water Assessment Tool (SWAT) model at Tarbiat Modares University, Tehran, Iran, 2022
- ❖ Workshop of modeling and forecasting climate changes using CMIP6 models at Kharazmi University, Tehran, Iran, 2023